

PC 504 AI	MACHINE LEARNING					
Pre-requisites			L	T	P	C
			3	-	-	3
Evaluation	CIE	40 Marks	SEE		60 Marks	

Course Objectives :	
1	To introduce the basic concepts of machine learning and range of problems that can be handled by machine learning
2	To introduce the concepts of instance based learning and decision tree induction
3	To introduce the concepts of linear separability, Perceptron and SVM
4	To learn the concepts of probabilistic inference, graphical models and evolutionary learning
5	To learn the concepts of ensemble learning, dimensionality reduction and Clustering

Course Outcomes :	
On completion of this course, the student will be able to :	
CO-1	Explain strengths and weakness of different machine learning techniques
CO-2	Select suitable model parameter for different machine learning technique
CO-3	Design & implement various machine learning algorithms to a wide range of real world applications
CO-4	Evaluate available learning methods to develop the research based solutions in different domains.

UNIT – I	
Introduction: Learning, Types of Machine Learning, Machine Learning Examples , Decision Tree Learning	
Chp 2 →	Concept learning: Introduction, Version Spaces and the Candidate Elimination Algorithm.
Chp 3	Learning with Trees: Decision Tree Learning, the Big Picture
Linear Discriminants: Learning Linear Separators , The Perceptron Algorithm , Margins	

UNIT – II	
Estimating Probabilities from Data, Bayes Rule, MLE, MAP	
Naive Bayes: Conditional Independence, Naive Bayes: Why and How, Bag of Words Logistic Regression : Maximizing Conditional likelihood , Gradient Descent	
Kernels: Kernalization Algorithm, Kernalizing Perceptron,	
Discriminants: The Perceptron, Linear Separability, Linear Regression	
Multilayer Perceptron (MLP): Going Forwards, Backwards, MLP in practices, Deriving back Propagation.	

UNIT- III

Support Vector Machines: Geometric margins, Primal and Dual Forms, Kernelizing SVM Generalization & Overfitting: Sample Complexity, Finite Hypothesis classes, VC Dimension Based Bounds

Some Basic Statistics: Averages, Variance and Covariance, The Gaussian, The Bias-Variance Tradeoff Bayesian learning: Introduction, Bayes theorem. Bayes Optimal Classifier, Naive Bayes Classifier.

Graphical Models: Bayesian networks, Approximate Inference, Making Bayesian Networks, Hidden Markov Models, The Forward Algorithm.

UNIT - IV

Model Selection & Regularization: Structural Risk Minimization, Regularization, k-Fold Cross validation

Linear Regression: Linear regression, minimizing squared error and maximizing data Likelihood

Neural Networks: Back Propagation,

Deep Neural Networks: Convolution, Convolution Neural Networks, LeNet-5 architecture

Boosting: Boosting Accuracy, Ada Boosting, Bagging

UNIT -V

Clustering: Introduction, Similarity and Distance Measures, Outliers, Hierarchical Methods, Partitional Algorithms, Clustering Large Databases, Clustering with Categorical Attributes, Comparison.

Dimensionality Reduction: Linear Discriminant Analysis, Principal Component Analysis Interactive Learning: Active Learning, Active Learning, Common heuristics, Sampling bias , Safe Disagreement Based Active Learning Schemes

Semi-Supervised Learning: Semi-supervised Learning, Transductive SVM, Co-training

Reinforcement Learning: Markov Decision Processes, Value Iteration, Q-Learning

Suggested Reading:

1 ✓	Tom M. Mitchell, Machine Learning, Mc Graw Hill, 1997
2 ✓	Christopher Bishop, Pattern recognition & Machine Learning, Springer 2006.
3 ✓	Stephen Marsland, Machine Learning - An Algorithmic Perspective, CRC Press, 2009.
4	Margaret H Dunham, Data Mining, Pearson Edition., 2003.
5	Galit Shmueli, Nitin R Patel, Peter C Bruce, Data Mining for Business Intelligence, Wiley India Edition, 2007
6	Rajjan Shinghal, Pattern Recognition, Oxford University Press, 2006.